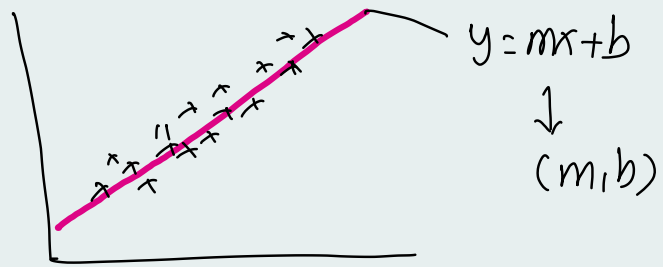


Linear Regression



(m, b) can be find using.

closed form
solution
(OLS)

non-closed form
solution
(Gradient Descent)

- OLS mean direct formula will be available.
- GD don't have any direct formula will use derivatives.
- If no of dimensions increases finding the direct formula is very difficult, that's the reason will go with GD (efficient)

OLS

$$y = mx + b$$

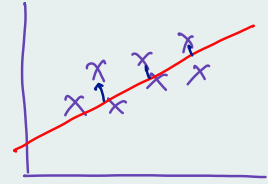
$$b = \bar{y} - m\bar{x}, \quad m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$\bar{x} \rightarrow \text{mean}(x)$

$\bar{y} \rightarrow \text{mean}(y)$

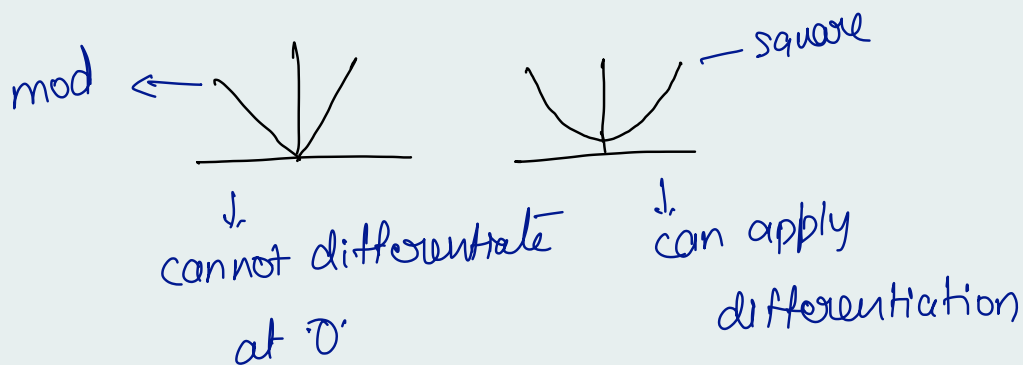
Implementation:

- Find the distance btw each point from line ($d_1, d_2, \dots, d_n$)



$$\begin{aligned}\text{Error} &= d_1 + d_2 + \dots + d_n \\ &= d_1^2 + d_2^2 + \dots + d_n^2\end{aligned}$$

Note: Reason for square is point can be above the line (+ve) or below the line (-ve)



$$\text{Error Function} = \sum_{i=1}^n d_i^2$$

↓
same fun can written in

d = diff btw actual vs predicted

$$d_i = (y_i - \hat{y}_i)$$

$$\boxed{E = \sum_{i=1}^n (y_i - \hat{y}_i)^2} \rightarrow \text{It should be minimum}$$

↓
How it is related with (m & b)

$$\downarrow \\ \hat{y} = mx + b$$

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

$x_i, y_i \rightarrow$ from data can not

$m, b \rightarrow$ can change change

Error function is depending on m, b

change in $m, b \rightarrow$ change in line

How much change in line with change in m & b ?

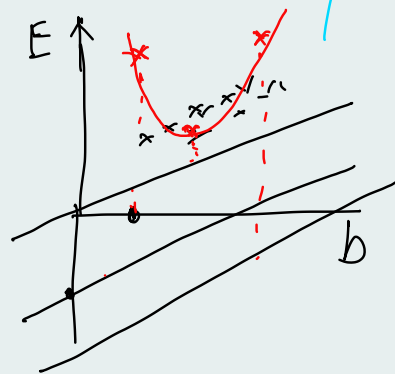
if $b=0$

$$\begin{aligned} E(m, 0) &= \sum_{i=1}^n (y_i - mx_i + b) \\ &= \sum_{i=1}^n (y_i - mx_i) \end{aligned}$$

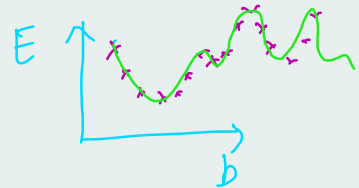
(line will always pass through origin)

if $m=0$

$$E(0, b) = \sum_{i=1}^n (y_i -$$



[making $b=0$, and draw multiple lines find error]



Partial Derivative on 'b'

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - mx_i - b)^2 = 0$$

$$\Rightarrow \sum \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$\Rightarrow \sum -2 (y_i - mx_i - b) = 0$$

$$\Rightarrow \sum (y_i - mx_i - b) = 0$$

$$\Rightarrow \frac{\sum y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum b}{n} = 0$$

equal to mean formula

$$\Rightarrow \bar{y}_i - m\bar{x} - b = 0$$

$$\Rightarrow b = \bar{y}_i - m\bar{x}$$

$\frac{b+b+\dots+b \rightarrow n \text{ times}}{n} \Rightarrow \frac{nb}{n} = b$

With respect to 'm'

$$E = \sum (y_i - mx_i - \bar{y} + m\bar{x}_i)$$

$$\frac{\partial E}{\partial m} = \sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x}_i)^2 = 0$$

$$\Rightarrow 2 \sum (y_i - mx_i - \bar{y} + m\bar{x}_i) (-x_i + \bar{x}_i) = 0$$

$$\Rightarrow -2 \sum (y_i - mx_i - \bar{y} + m\bar{x}_i) (x_i - \bar{x}_i) = 0$$

$$\Rightarrow \sum (y_i - mx_i - \bar{y} + m\bar{x}_i) (x_i - \bar{x}_i) = 0$$

$$\Rightarrow \sum [(y_i - \bar{y}) - m(x_i - \bar{x}_i)] (x_i - \bar{x}_i) = 0$$

$$\Rightarrow \sum [(y_i - \bar{y})(x_i - \bar{x}_i) - m(x_i - \bar{x}_i)^2] = 0$$

$$\Rightarrow \boxed{m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}_i)}{(x_i - \bar{x}_i)^2}}$$

Multiple Linear Regression:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

100 $\rightarrow$ no. of rows
n - no. of columns

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \dots & \beta_n x_{1n} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \dots & \beta_n x_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ \beta_0 & \beta_n x_{1001} & \beta_2 x_{1002} & \dots & \beta_n x_{100n} \end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix}}_{A/X} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}}_{B/\beta}$$

$$\hat{y} = (A B)^T \Rightarrow \hat{y} = (X B)^T$$

Let's write generic equation with 'n' rows & 'm' columns.

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{12} & \beta_2 x_{12} & \beta_3 x_{13} & \dots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \beta_3 x_{23} & \dots & \beta_m x_{2m} \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ \beta_0 & \beta_m x_{n1} & \beta_2 x_{n2} & \beta_3 x_{n3} & \dots & \beta_m x_{nm} \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & x_{23} & \dots & x_{2m} \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n3} & \dots & x_{nm} \end{bmatrix}}_{[n \times m]} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}}_{[m \times 1]}$$

$\rightarrow$ If we multiply A with B will get same previous matrix.

$$\hat{Y} = X\beta$$

$X \rightarrow$ total data set with 1 at the end.

$$\text{error}(e) = Y - \hat{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$\hat{y} \rightarrow$ predicted score

$\beta \rightarrow$ no. of feature coeff.

From simple LR:

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ we can write it as } E = e^T e.$$

$E = e^T e$. (If we multiply e^T & e , will get same

$$= \begin{bmatrix} (y_1 - \hat{y}_1) & (y_2 - \hat{y}_2) & \dots & (y_n - \hat{y}_n) \end{bmatrix} \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

$$= (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_n - \hat{y}_n)^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\therefore E = e^T e = (Y - \hat{Y})^T (Y - \hat{Y})$$

$$= (Y^T - \hat{Y}^T) (Y - \hat{Y})$$

we know

$$(A+B)^T = A^T + B^T$$

$$(A-B)^T = A^T - B^T$$

$$[\text{we know } \hat{Y} = X\beta]$$

$$= (Y^T - (X\beta)^T) (Y - X\beta)$$

$$= YY^T - \underline{Y^T X \beta} - \underline{(X\beta)^T Y} + (X\beta)^T (X\beta)$$

$y^T x \beta$ & $(x\beta)^T y$ are equal. Let's prove.

Let's say $y = A$, $x\beta = B$.

$$\therefore A^T B = B^T A \quad \text{Let's prove it} \\ \text{--- (1)}$$

$$(A^T B)^T = (A^T)^T \cdot B^T \\ = A \cdot B^T \text{--- (2)}$$

from (1) & (2).

$$A^T B = (A^T B)^T$$

$$A^T B = C.$$

$$C = C^T.$$

$$\therefore (y^T x \beta) = (y^T x \beta)^T$$

$$y = \begin{bmatrix} \equiv \\ \equiv \\ \equiv \\ \equiv \end{bmatrix}_{n \times 1}$$

$$y^T = 1 \times n$$

$$\therefore \left[(1 \times n) \cdot (n \times (m+1)) \cdot ((m+1) \times 1) \right]$$

$$= \left[(1 \times (m+1)) ((m+1) \times 1) \right]$$

$$= 1 \times 1$$

$\therefore$ The above equation is re written as

$$E = y^T y - 2y^T x \beta + \beta^T x^T x \beta.$$

To get the min 'β'

$$\frac{dE}{d\beta} = \frac{d}{d\beta} [y^T y - 2y^T x \beta + \beta^T x^T x \beta] = 0.$$

$$\left[0 - 2y^T x + \frac{d}{d\beta} [\beta^T x^T x \beta] \right] = 0$$

Another way to derive Θ (w.r. to m): using matrix Inverse.

1. Define a model

$$y = Xw + \epsilon$$

m-rows
n-cols.

where X is an $m \times n$ matrix
 w is $n \times 1$ vector of wts
 y is $m \times 1$ target vector
 ϵ is error vector

2. Define Cost Function.

$$\begin{aligned} L &= (y - \hat{y})^2 \\ &= (y - Xw)^2 \\ &= (y - Xw)^T \cdot (y - Xw) \end{aligned}$$

[$\because$ dot product]

3. Expand the cost Function.

$$\begin{aligned} L &= y^T y - y^T Xw - X^T w^T y + X^T w^T Xw \\ &= y^T y - 2y^T Xw + X^T w^T Xw \end{aligned}$$

4. Compute the Gradient w.r. to w .

$$\frac{dL}{dw} = \frac{d}{dw} (y^T y - 2y^T Xw + X^T w^T Xw)$$

$$= 0 - 2y^T x + 2x^T x w$$

set the gradient to zero to minimize loss.

$$- 2x^T y + 2x^T x w = 0.$$

$$x^T x w = x^T y$$

$$\boxed{\therefore w = (x^T x)^{-1} (x^T y)}$$

This is the closed form solution for OLS.

OLS stats model:

Df Residuals: $n - k - 1$

no. of rows - no. of cols - 1

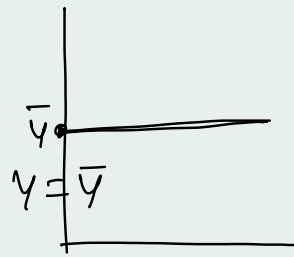
DF model: no. of columns

Covariance Type: non robust

- Explains the correlation btw the variables.

F-Stats:

let's say $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$



— F-test of overall significance indicates, whether your LR model provides a better fit to the data than a model with no-indp variables means only with the intercept.

— we are calling it an intercept only model.

H_0 = model with no indp variables fit the model well

H_1 = model fits the data better than the intercept-only model.

let's say fit the model with only intercept

got $MSE = 200$

fit with indp var, if MSE is changing.

rejecting H_0 .

Log-likelihood.

— The value should be high

— need to check with one col at a time.

what ever the combo of x_1, x_2, x_3 are having

high log-likelihood value. Those combo is good.

- It is compare coeft values of each variables in the process of creating the model.
- Log-lik value is a measure of goodness of fit. values ranging from $[-\infty$ to $+\infty]$

AIC & BIC : Both will the accuracy of model.

$$AIC = -2 \log L_i + 2P_i$$

$$BIC = -2 \log L_i + P_i \log n$$

std.err:

- It is the std.err of coeft.
- Use of this std.err is to find t-test value.

$$t\text{-value} = \frac{\text{coeft}}{\text{std.err}}$$

$t & p > |t|$

t-test p value will help us to understand how mch change in 'y' w.r. to $x_1, x_2, x_3 \dots$

H_0 : variables has no correlation with target var.

H_1 : Ind var has corr with target var.

if $p < 0.05$:

reject null hyp.

else:

fail to reject H_0 .

Omnibus & Prob(Omnibus)

— This will tell about the normality of Residuals.

$$\text{Residuals} = y - \hat{y}$$

— If prob(Omnibus) value is close to '0' means residuals follow normal dist.

Durbin-Watson

This test is for checking the autocorrelation in residuals.

if $DW \approx 2$ (no autocorr)

if $0 < DW < 2$ (+ve auto corr)

if $2 < DW < 4$ (-ve auto corr)

Jarque-Bera & p(JB)

— To test Residuals are normally dist.

Assumptions of LR

1. Linear Relationship btw input & output
2. NO multicollinearity
3. Normality of Residuals
4. Homoscedasticity
5. NO autocorrelation of errors.

1. Linear Relationship:



Linear



Linear



no-linear.

— each input var has linear relationship with output

2. Multicollinearity:

x_1 x_2 x_3 y .

— ~~no~~ should not have any relation btw x_1, x_2, x_3 .

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

— Here β_1 will explain about how much change

in 'y', when will change in x_1 , where

x_2 & x_3 are const.

— same for x_2 & x_3 also.

— But when x_1 is changing x_3 are affecting means

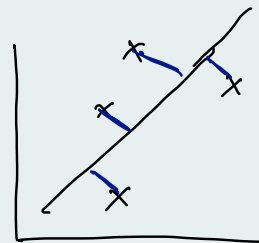
there is a corr btw x_1 & x_3 .

Technique to find is $VIF = \frac{1}{1-R^2}$.

3. Normal Residuals:

$$\text{Residual} = y - \hat{y}$$

diff btw actual vs predicted

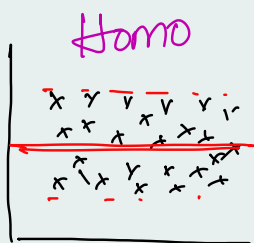


— take the each error (distance) & plot it should follow normal dist where mean of residuals should be zero.

4. Homoscedasticity:

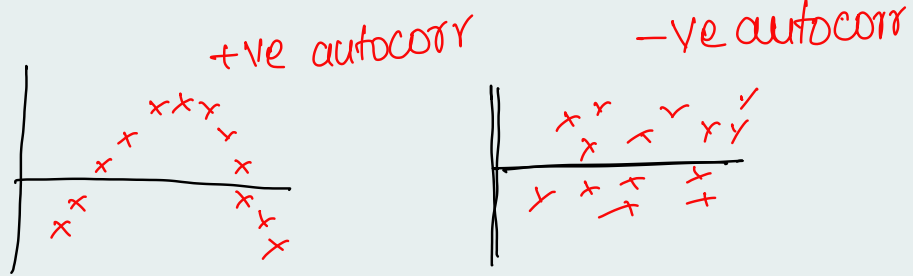
Homo = same

scasticity = spread / scatter.



- means should have some scatter

5. No Autocorr of errors



- no relationship btw the errors.

